



Experimental investigation on temperature-dependent effective thermal conductivity of ceramic fiber felt

Fei Wang, Yinan Wang, Chuang Sun^{*}, Pei Zhang, Xinlin Xia

School of Energy Science and Engineering, Harbin Institute of Technology, Harbin, Heilongjiang, 150001, China

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ABSTRACT

This paper presents a method to invert the effective thermal conductivity of ceramic fiber felt under high temperatures using experimentally measured specific heat capacity, density, and transient temperature data. The method combines the solution of the heat transfer equation with an inversion method based on real-valued genetic algorithm (GA) optimization, and 11 target parameters can be simultaneously inverted with one experimental dataset. The effective thermal conductivities of ceramic fiber felt were obtained by inversion under ambient pressures of 2 Pa and 100 kPa, compression ratios of 0.8–1.0, and temperature ranges of 295–1273 K. The results indicate that the effective thermal conductivities of ceramic fiber felt derived from the inversion method are in concurrence with the outcomes of the steady state method. The effective thermal conductivities of the ceramic fiber felt range from 0.069 to 0.321 W m⁻¹ K⁻¹. It is observed that the effective thermal conductivity increases with higher temperature, and decreases with growing compression ratio. The impact of ambient pressure on effective thermal conductivity is linked to the level of compression.

1. Introduction

The rapid progress of aerospace technology has led to the creation of hypersonic vehicles that operate at flight speeds greater than Mach 5 and for longer durations in the atmosphere. Despite such advancements, the vehicle's wind-facing surface and wing leading edge are vulnerable to elevated aerodynamic heating, with the thermal conditions being exceptionally severe [1,2]. The temperature at the nose cone reaches up to 1793 °C, and the leading edge of the wing can hit as high as 1455 °C [3] when the aircraft is flying at Mach 8 in the atmosphere. A reliable thermal protection system is necessary to ensure that the internal instruments and equipment function within the suitable temperature limits.

Aerogel materials are frequently utilized as thermal insulation materials due to their low density and thermal conductivity [4,5]. However, aerogel material exhibits low mechanical properties and brittleness. Additionally, the internal nanopore structures of aerogel materials readily collapse at high temperatures, creating issues in meeting the requirements of extreme environmental applications including thermal shock and long-time exposure to high temperatures in aerospace and other fields [6]. In contrast, ceramic fiber materials offer the advantages such as high-temperature resistance, antioxidant

properties, chemical stability, and load bearing capacity. They can be used as an aerogel reinforcing agent and as an infrared light screening agent. Additionally, they can serve solely as an insulating material at high temperatures. Due to their beneficial properties, ceramic fiber materials find widespread application in industries such as aerospace, chemo-metallurgy, and nuclear power plants [7].

Ceramic fiber materials with high porosity are typically utilized in atmospheric conditions, with the gaps between fibers being filled with gas. Due to inadequate gas pressure differences and the unconnected nature of the gaps between the fibers within the material, convective heat transfer in the thermal insulation materials with micro and nano-scale pores can be disregarded [8,9]. Thermal transfer mechanisms in ceramic fiber materials involve heat conduction of gases, solid heat conducting, and radiation heat transfer. Gas heat conduction in ceramic fibers includes two processes. Technical terms will be explained when first used. First, gas molecules in pores collide with each other, resulting in heat transfer or gas thermal conductivity. Second, gas molecules collide with the solid surface of the fiber, resulting in heat transfer or gas-solid thermal conductivity. For inorganic ceramic fibers, phonons are the main mechanism in solid heat transfer, since phonon motion properties in solid materials determine thermal conductivities. Radiative heat transfer occurs when an object with a temperature above

^{*} Corresponding author.

E-mail address: sunc@hit.edu.cn (C. Sun).

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absolute zero emits infrared radiation. Thermal transfer of radiation depends mainly on the optical response of materials to electromagnetic waves in the region of infrared region. Therefore, obtaining the effective heat conductivity of ceramic fiber felt is crucial for designing and evaluating heat protection structures in practical applications.

Numerous theoretical models exist for forecasting the effective thermal conductivity of ceramic fiber based on their compositions, fractions, porosity, and thermophysical parameters. Theoretical modeling by Arambakam [10] at a temperature of 573 K calculated the effect of fiber diameter on the effective thermal conductivity of ceramic fiber. The results indicate that micro and nanofibers have low transmittance in infrared and low thermal conductive radiant, particularly at high temperatures.

Experimental techniques have been extensively utilized to measure the effective heat conductivity of ceramic fiber felt. Nonetheless, their utilization in high-temperature applications is constrained. The two primary methods for measuring thermal conductivity to date are the steady state method [11–14] and the nonequilibrium method [15–19]. In steady state methods, a temperature gradient is applied to the test sample and thermal conductivity is measured directly from the heat flux density [20]. Gibson [21] experimentally verifies the effective thermal conductivity of high loft nanofiber battings comprising carbon fiber and various polymeric materials using the hot-air method. The temperature of the top and bottom plates is 50 °C and 10 °C, respectively. Wang et al. [22] conducted a study using a hotplate methodology to measure the thermal conductivity of ZrO₂ fibers at 300, 500, 700, and 900 °C. The steady state method was utilized to obtain the effective thermal conductivity of the specimen at the average temperature of each temperature range. It should be noted that this measurement procedure is specifically relevant to the specimen value in the concerned temperature range. Zhu et al. [23] conducted thermal conductivity tests on nanosheet papers made of cellulose and 5% boron nitride at room temperature. Diaz, J. et al. [24] used a steady state bridge method with aluminum radiation shielding and vacuum enclosure mounting to measure thermal conductivity in films of cellulose nanocrystals (CNCs) near room temperature. The steady state measurement techniques pose certain drawbacks. For instance, the characteristic temperature that aligns with the thermal conductivity measurements is the average of the temperatures on the top and bottom surfaces of the sample. Regrettably, this approach is generally suitable for materials with linear thermal conductivity changes with temperature [14,25,26]. Additionally, this method necessitates prolonged wait times for the sample to attain a steady state. Moreover, precise measurement of temperature difference is crucial as it poses a challenge in determining the thermal conductivity of thin or high-conductivity materials at low temperature differentials [20].

Non-equilibrium methods generate temperature fluctuations on the surface and detect temperature response on the rear side of the specimen to measure thermal conductivity k or diffusivity α [27], in comparison to steady state methods. The measurement methods significantly limit specimen morphology and heat flux direction for non-equilibrium methods. Various methods were utilized to measure the thermal conductivity of fibrous materials. These include the transient planar source (TPS), Joule heating using an alternating current (AC), temperature wave analysis, laser flash technique, and periodic laser spot heating with radiation thermometry. TPS is commonly utilized in thermal conductivity testing. The Hot Disk Thermal Analyzer is a typical device which operates based on the TPS method. To conduct measurements, various types of sensors such as Kapton, mica, and Teflon sensors are required, depending on the desired measurement temperature. Jia et al. [28] measured the thermal conductivity of SiO₂–Al₂O₃ ceramic sponge composite (SAC sponges) in the range of 20–120 °C using the Hot disk thermal conductive analyzer with the method of transient planar springs. Similarly, Su et al. [29] determined the thermal conductivity test of SiC ceramic nanowire aerogel at temperatures up to 600 °C in N₂

using the Hot Disk TPS instrument with a 9.7 mm mica sensor. Nevertheless, the use of mica sensors may overestimate the effective thermal conductivity of low thermal conductive materials (less than 1.45 W·m⁻¹·K⁻¹) because of non-negligible heat losses at the sensor's side [30].

Heat transfer is the process of temperature diffusion, where the diffusion coefficient is referred to as thermal diffusivity α . Non-equilibrium methods generally rely on measuring α , and the thermal conductivity of a specimen is determined by the product of thermal diffusivity, heat capacity, and material density. Researchers, including Hashimoto et al. [31], have measured thermal diffusivity in the temperature range of 20–200 °C for polymer films using the Joule heating method and the temperature wave method. Morikawa, J. and Hashimoto, T. [32], measured the thermal diffusivity of a paper using a porous and thin AC Joule heating method at room temperature. Hashimoto, T. et al. [33] initially developed temperature wave analysis (TWA) to obtain the frequency-dependent thermal diffusivity as a function of temperature. Similarly Kojiro, U. et al. [34] used a thermal wave analyzer to measure the thermal diffusivity of the nano papers at room temperature. Shimazaki, Y. et al. [35] determined the thermal diffusivity of transparent cellulose nanofiber/epoxy resin nanocomposites in the thickness direction at room temperature using thermal wave analysis. In contrast, the AC calorimetric method was employed to measure the thermal diffusivities parallel to the surface of the sample. Kojiro, U et al. [36,37] conducted the thermal diffusivity tests on nonwoven nitrocellulose sheets near room temperature and the optically transparent flexible films with surface-exposed nanocellulose skeletons in thickness directions at temperatures ranging from 25–110 °C using a periodic heating method. Additionally, Kojiro, U. et al. [38] utilized the flash method and the periodic heating method to measure the thermal diffusivity of carbon fiber fillers below 100 °C. Liu et al. [39] used the periodic laser spot heating radiation thermometry method to measure the thermal diffusivity of SiC fibrous mats from room temperature to 600 °C. Subsequently, they calculated the mats' thermal conductivity. Yves J. et al. [40] introduced the method (4 L) for estimating the transfer function to the center of a symmetrical stack to measure the thermal diffusivity of both opaque insulating material and semi-transparent insulating materials at high temperatures. Meanwhile, Zeng et al. [41] utilized the lasers flash method to measure the in-plane thermal diffusivity of a combination of boron nitride nanotubes and cellulose nanofibers at 30 °C. Pretreatment, including black-coating or metal-sputtering process, is required for the specimen to undergo laser flash method and periodic heating method testing to improve accuracy. Moreover, thermal conductivity calculations based on an indirect approach utilizing diffusivity measurement methods involve a level of uncertainty. It is evident that the non-equilibrium method yields an effective thermal conductivity below 1000 K. Moreover, transient measurement allows for obtaining effective thermal conductivity at a single temperature per instance. To obtain the temperature-dependent thermal conductive function, multiple measurements are required at various temperatures, which must be averaged. Additionally, the ceramic fiber felt generally possesses excellent thermal insulation properties. However, its interwoven fiber structure complicates the task of ensuring temperature uniformity during experimentation.

The study combines the heat transfer equation and the inversion process of a real-values based genetic optimization to simultaneously invert the effective thermal conductivity of the sample at 11 characteristic temperatures ranging from 295 K to 1273 K by using the transient temperature of the ceramic fiber felt. Additionally, the study explores the effects of the compression rate and the ambient pressure on the effective thermal conductivity of the ceramic fiber felt. Compared to steady state measurements, the transient inversion method presented in this paper demonstrates greater efficiency, accuracy, and reliability.

2. Experimental procedure

2.1. Specimen and structure characterization

The ceramic fiber felt samples investigated in this paper are mainly composed of alumina fibers, the uncompressed ceramic fiber felt measures $300 \times 300 \times 40$ mm. It can be seen from Fig. 1(b) that the fibers in the samples do not have a regular distribution pattern, and the fibers are crossed with each other longitudinally and transversely to form giant pores. The small diameter of the sample fibers ranges from $1 \mu\text{m}$ to $10 \mu\text{m}$, with minimal fiber-to-fiber contact, which contributes to the improved thermal insulation performance of the sample. Meanwhile, ceramic fibers felt are highly sensitive to moisture as water vapor adheres to the surface of the fiber rods. This impacts the sample's effective thermal conductivity. Consequently, the samples undergo supercritical drying in a high-temperature vacuum chamber to eliminate the water vapor before conducting experiments.

The heat transfer process of the sample includes three physical parameters: density, specific heat capacity, and thermal conductivity. Accurately measuring the sample's specific heat capacity and density is necessary to calculate its thermal conductivity using the inversion algorithm. Liu et al. [42] indicates that the density and specific heat of the porous material can be estimated from the components and their respective densities and specific heats, and thus the specific volumetric heat of ceramic fiber experienced can be expressed as

$$c_{v,\text{felt}}(T) = (1 - \varphi)c_{v,\text{fiber}}(T) + \varphi c_{v,\text{gas}}(T) \quad (1)$$

where, $c_{v,\text{felt}}(T)$, $c_{v,\text{fiber}}(T)$ and $c_{v,\text{gas}}(T)$ are the volumetric specific heat of the present ceramic fiber felt, ceramic fiber rods, and nitrogen, respectively. φ is the porosity of the specimen. The volumetric specific heat of ceramic fiber rods is determined from its density and specific heat reported in the literature. The specific heat of the sample measured in this study calculated from Eq. (1) at the temperature range from 300 K to 1300 K is polynomial fitted as a function of temperature as

$$c_p(T) = 0.000765T^2 - 0.73424T + 841.591409 \quad (2)$$

Additionally, we utilized the DSC (Netzsch DSC 404 F3, Germany) to test the specific heat of ceramic fiber felt when uncompressed within the temperature range 323 K–1473 K. As illustrated in Fig. 2, the experimental data and fitting formula displayed strong consistency, confirming the precise model for this sample's specific volumetric heat.

The compression ratio is determined based on the compressive endurance of the ceramic fiber felt, which is defined as the ratio of the compressed volume to the original specimen volume. Since the volume of ceramic fiber felt varies with the thickness direction, the compression ratio of the current study can be expressed as follows

$$\varepsilon = \frac{V_c}{V} = \frac{L_c}{L} = \frac{\rho}{\rho_c} \quad (3)$$

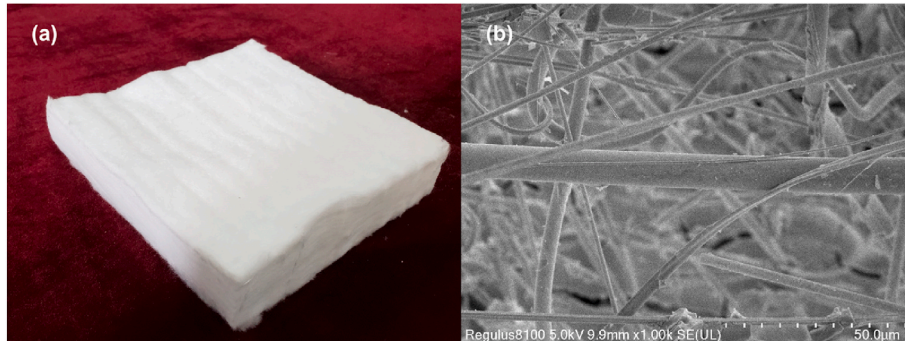


Fig. 1. (a) Photographs and (b) scanning electron micrographs (SEM, Regulus8100, Hitachi, Japan) of ceramic fiber felt specimen.

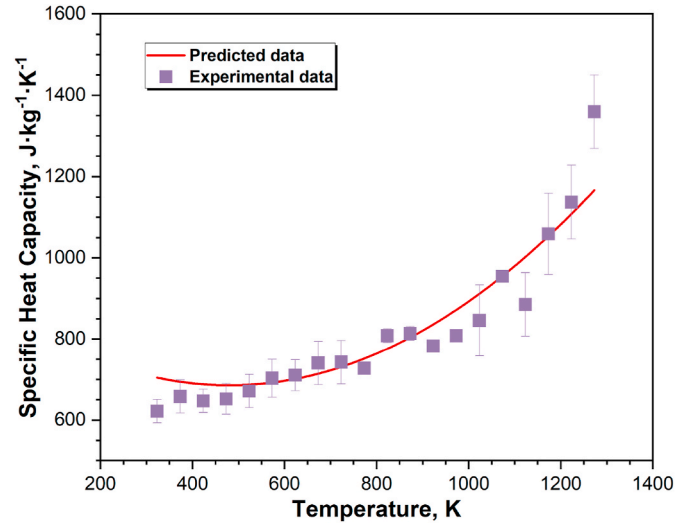


Fig. 2. Specific heat of present ceramic fiber felt.

where V_c , L_c , ρ_c and are the volume, thickness, and density of compressed specimen, respectively; V , L and ρ are the volume thickness and density of uncompressed specimen, respectively; the subscript c presents the compressed specimen. The relationship between porosity and compression rate is

$$\varphi_c = 1 - \frac{1 - \varphi_o}{\varepsilon} \quad (4)$$

where φ_c is the specimen porosity corresponding to compression rate ε , and φ_o is the porosity of uncompressed specimen.

2.2. High-temperature test load test platform

Experimental measurements in current studies are shown in Fig. 3. This experiment is carried out by proving a radiation heat source through highly resistive slot graphite, which can be ramped up to a maximum of 2300 K in a vacuum or shielded gas atmosphere, e.g., argon, nitrogen. A type C thermocouple (NANMAC, USA) is used to measure the furnace temperature, and a proportional-integral-derivation (PID) controller is used for automatically adjusting the radiation heater power by comparing the thermocouples feedback signal with the experimental set-up temperature to achieve temperature control.

The specimen to be tested is placed between two identical graphite plates forming a sandwich structure, and the graphite plate are $300 \times 300 \times 10$ mm in size. A high-temperature insulating chamber is utilized to position the sandwich structure. One side of the outer surface (the hot surface) is directly heated by a radiant heater, while the other side of the

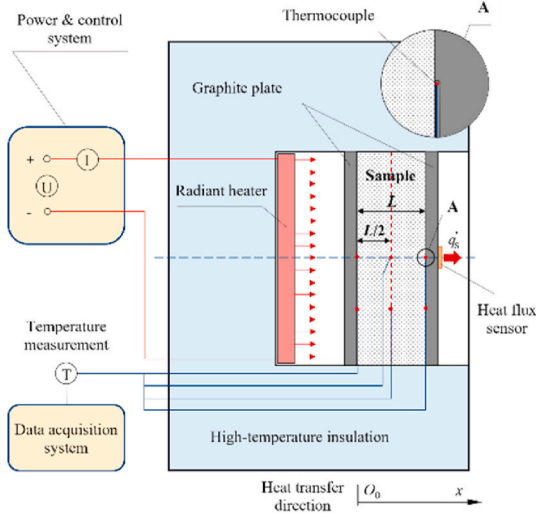


Fig. 3. Schematic of the experimental measurements as well as the coordinate system.

outer surface (the cold surface) cools naturally. The sandwich construction is enclosed by the high-temperature insulation material. To protect that the device and sample are protected from oxidation at high temperatures, the high-temperature insulation chamber is either vacuum-sealed or filled with shielding gases. The pressure within the chamber can be set within a range of 0.01 Pa–200 kPa, with automatic feedback provided by a pressure sensor.

Fig. 4 displays the thermocouple arrangement for testing the samples. To collect temperature data from various sample locations, six type K thermocouples (OMEGA, USA) with an effective range of 273 K–1473 K are utilized to capture instantaneous temperature readings. Specifically, three of the thermocouples (TC1, TC3, and TC5) are situated at the sample's geometrical center while the other three (TC2, TC4, and TC6) are located 75 mm away from the geometric center. The thermocouples are arranged along the sample thickness on the heated (TC1, TC2), intermediate (TC3, TC4), and cold (TC5, TC6) surfaces, with a USB data acquisition module connecting to the thermocouples to import temperature data into the computer system. Due to the temperature differences between the locations of TC1 and TC2 and TC3 to TC6, different thermocouple models were used. Model CHAL-015-BW, with a wire diameter of 0.015 inches, was used for the higher temperature locations, while model CHAL-010-BW, with a wire diameter of 0.010 inches, was used for the lower temperature locations. A heat flux meter

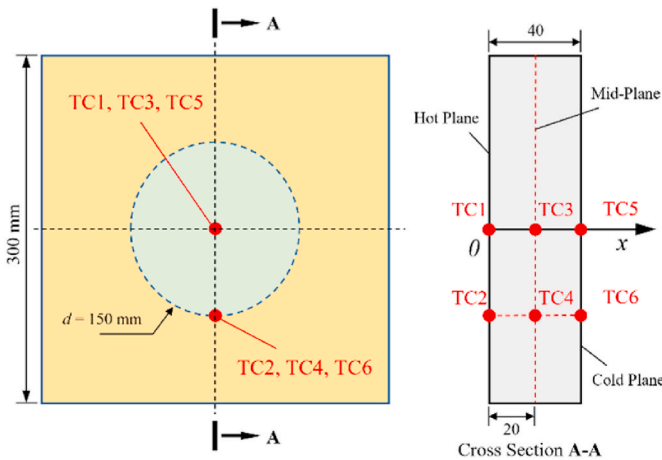


Fig. 4. Location of different thermocouples on the ceramic fiber felt sample.

sensor (DaqPRO 5300, Fourier, Israel) is placed on the top of the sandwich construct's cold surface for monitoring.

To ascertain the proper connection of components and precise measurement of thermocouples, inspection of the temperature data acquisition system before experimentation is necessary. Two fixed point thermometric temperatures, the ice and water mixture, and boiling water, are used for calibration of thermocouples. The results indicate that the measurement error of thermocouples is within $\pm 0.5^\circ\text{C}$. Calibrating thermocouples in the high-temperature range of 500–1300 K was achieved by employing a blackbody source (LAND P1200B) with 200 K intervals. The temperature difference between thermocouple readings and standard values was contained within 0.75% of the error bound.

2.3. Experimental validation of 1D heat transfer

To simplify the problem, we assume that heat transfer in the thickness direction of the sample is 1D inside a central circular area with a 75 mm radius, as shown in Fig. 4. The aerogel is selected for testing this assumption, as it demonstrates better thermal insulation performance compared to the ceramic fiber felt. Consequently, the former can generate a greater temperature gradient within the specimen. In addition, the size of the aerogel samples and the arrangement of thermocouples are identical to those of the ceramic fiber felt. Data from thermocouples TC1/TC2, TC3/TC4, and TC5/TC6 are compared in Fig. 5. The TC1 and TC2 thermocouples are located at the center and periphery of the heated surface, while the TC3 and TC4 thermocouples measure the midplane temperature at the middle and periphery regions of the circular zone. The TC5 and TC6 thermocouples are situated in the center zone and periphery of the cold plane, respectively. The furnace temperature is 1600 K and the ambient pressure is 100 kPa.

During the experiment, it was observed that the temperatures measured by TC1/TC2, TC3/TC4, and TC5/TC6 were quite similar, with a maximum temperature difference of around 6 K, 5 K, and 4.5 K, respectively. Additionally, the temperature differences between the heated-middle and middle-cold surfaces were approximately 550 K and 650 K, respectively, which were two orders of magnitude larger than the maximum temperature differences recorded by TC1/TC2, TC3/TC4, and TC5/TC6. As a result, the heat transfer in the central circular zone can be approximately considered 1D.

3. Heat transfer model and inverse method

3.1. Direct problem

3.1.1. Combined conductive and radiative heat transfer

The heat transfer mechanism in ceramic fiber felt includes heat conduction and radiation heat exchange between the solid phase materials, heat conduction, and convection in the gas phase within the pore, as the material pores are comparable to the free range of gas molecules, so that convection can be neglected. Therefore, the stationary heat conduction-radiation coupled heat transfer equation within the thermal insulation felt is given by the following equation [43,44].

$$\rho(T)c_p(T)\frac{\partial T}{\partial \tau} = \nabla \cdot (k_c(T)\nabla T) - \nabla \cdot \vec{q}_r \quad (5)$$

Where $k_c(T)$ is the thermal conductivity consists of the contributions of solid conduction, gas conduction, and solid-gas coupling conduction, $\vec{q}_r$ is the radiation heat flux.

3.1.2. Rosseland diffusion approximation

For an optically thick medium, the radiated energy decays over a very short distance before its effect can propagate to the faraway position. Then, radiation can be considered as a diffusion process, and the total heat flux $\vec{q}_r$ can be expressed as the sum of the conductive $\vec{q}_c$ and

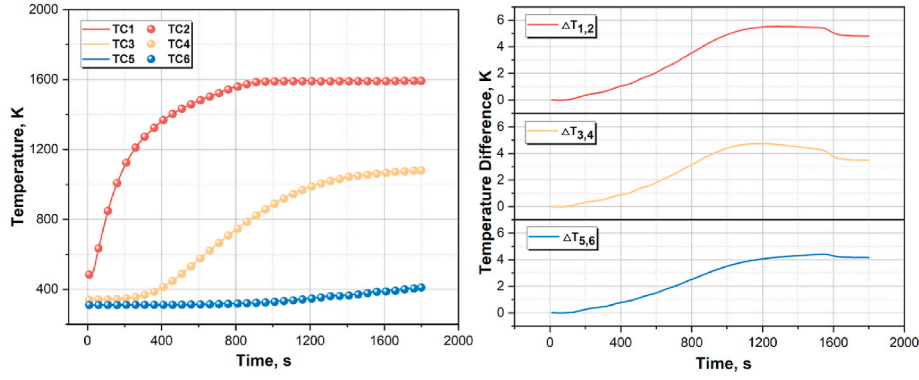


Fig. 5. Experimentally measured heated surface temperature $T_H(\tau)$, midplane temperature $T_M(\tau)$ and cold surface temperature $T_C(\tau)$ of ceramic fiber felt at different positions for the furnace temperature of $T_f = 1600$ K and pressure of $p = 100$ kPa.

radiative $\vec{q}_r$ heat fluxes given by Fourier's law [43,44]

$$\vec{q}_t = \vec{q}_c + \vec{q}_r = -k_c(T)\nabla T - k_r(T)\nabla T = -k_{\text{eff}}(T)\nabla T \quad (6)$$

where $k_c(T)$ and $k_r(T)$ are the conductive and radiative thermal conductivities, respectively. The effective thermal conductivity $k_{\text{eff}}(T)$ can be expressed as the sum of the conductive and radiative thermal conductivities [43,44]

$$k_{\text{eff}}(T) = k_c(T) + k_r(T) \quad (7)$$

in this study, the ceramic fiber felt is considered as an optically thick medium due to the strong attenuation of radiation, and the Rosseland diffusion approximation (RDA) is used to solve the radiation transfer equation. The radiative thermal conductivity $k_r(T)$ can be expressed as [43,44]

$$k_r(T) = \frac{16\sigma n^2 T^3}{3\beta_R^{\text{tr}}(T)} \quad (8)$$

where, σ is the Stefan-Boltzman constant ($\sigma = 5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$), n is the refractive index, $\beta_R^{\text{tr}}(T)$ is the Rosseland-mean extinction coefficient calculated from Ref. [44].

3.1.3. Boundary and initial conditions

The boundary and initial conditions for the heat transfer process of the specimen are detailed as follows.

$$T(0, \tau) = T_H(\tau), \text{ and } T(L, \tau) = T_C(\tau) \quad (9)$$

$$T(x, 0) = T_0 \quad (10)$$

where $T_H(\tau)$ and $T_C(\tau)$ are the measured temperature at positions of $x = 0$ (heated surface) and $x = L$ (cold surface), respectively, and $T_0 = 295$ K is the initial temperature measured at the beginning of the experiment.

3.2. Inverse optimization method

The accuracy of the thermal conductivity data obtained through the optimization algorithm is crucial in solving the inversion problem. It is essential to incorporate the algorithm-generated thermal conductivity into the positive problem along with the boundary conditions in Eqs. (9) and (10). This will enable the calculation of temperature data at the sample's intermediate interface. By comparing this with the actual temperature of the experimental intermediate interface, the accuracy of the generated thermal conductivity data can be evaluated. In this context, a fitness function (defined as Eq. (11)) we can be constructed to assess the level of concordance between the temperatures measured and calculated.

$$F = \frac{1}{N} \left[\sum_{i=1}^N \sqrt{\left(\frac{T_{M,\text{exp}}(i) - T_{M,\text{pred}}(i)}{T_{M,\text{exp}}(i)} \right)^2} \right] \quad (11)$$

where N is the number of temperature sampling points, $T_{M,\text{exp}}(i)$ and $T_{M,\text{pred}}(i)$ are the i th temperature of midplane measured from experiment and calculated from the direct problem, respectively. From Eq. (16), it is evident that the greater the accuracy of the thermal conductivity values obtained from the inversion, the smaller the fitness function should be. In the present study, the temperature sample points start from the 100 s moment, and the temperature values are collected every 100 s interval until 10000s, and therefore $N = 100$ and the value is [1, 100].

Since the functional form of the $k_{\text{eff}}(T)$ is uncertain, a set of temperature values T_m , $m = 1, 2, \dots, 11$ is determined at 100 K intervals between 295 K and 1273 K, which corresponds to a thermal conductive of k_m , $m = 1, 2, \dots, 11$, respectively. The thermal conductivity at the remaining temperatures is determined according to linear interpolation, and the inversion of these 11 parameters is performed simultaneously in the following.

Gene algorithms are self-organizing, adaptive, and probabilistic heuristic search algorithms among many optimization-solving algorithms. Despite its lower search efficiency compared to other traditional algorithms based on gradient computation, the genetic algorithm is advantageous for multi-parameter intelligent optimization due to its global optimization ability and robustness to complex temperatures. Therefore, this paper employs the genetic algorithm to identify and solve multi-physics parameters in a coupled radiation-conductivity heat transfer process. The basic flowchart of the algorithm is shown in Fig. 6. In case of a fitness function F value less than 4×10^{-4} or generation reaching the maximum value of 200, the fitness function is considered to converge and the corresponding set of is considered as the output.

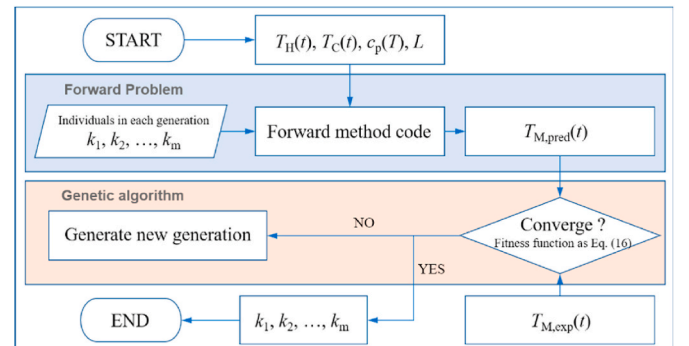


Fig. 6. Flow chart of the calculation process.

3.3. Sensitivity analysis

Inverse optimization computation sensitivity analysis is crucial because it can out reflect internal characteristics of the inverse problem and can provide insights into experimental design. The sensitivity coefficient for the parameter k_m ($m = 1, 2, \dots, 11$) is defined as

$$\chi(k_m, \tau) = \frac{\partial T_c(k_m, \tau)}{\partial k_m} \cdot \frac{k_m}{T_c(k_m, \tau)} \quad (12)$$

$$= \frac{T_c(k_m + k_m \Delta, \tau) - T_c(k_m, \tau)}{k_m} \cdot \frac{k_m}{T_c(k_m, \tau)}$$

where Δ is a small percentage change of parameter k_m set as $\Delta = 0.5\%$. The sensitivity coefficient in this study denotes the alteration in temperature at the sample's intermediate interface attributable to thermal conductivity perturbation $k_m \Delta$. A sensitivity coefficient near zero indicates that altering the thermal conductivity of the sample at a specific temperature in the heat transfer model lacks a significant impact on temperature. In some cases, the temperature may not change at all. When this happens, it becomes impossible to acquire the thermal conductivity value at the corresponding temperature based on temperature changes observed during the inversion process. Therefore, a sensitivity coefficient that is too small is unfavorable for the inversion process.

Since the materials in literature [45] have similar composition and physical properties to those used in this study, the effective thermal conductivity (shown in Fig. 7 (b)) predicted by the numerical method in literature [12] and the corresponding experimental temperature (shown in Fig. 7 (a)) is used to illustrate the sensitivity analysis in Fig. 7 (c). From Fig. 7(c), it is evident that the sensitivity coefficients for thermal conductivity decrease as the sample part's overall temperature rises with time. Additionally, when the middle surface of the sample part surpasses a certain temperature, the corresponding thermal conductivity loses its effectiveness, and the sensitivity coefficient becomes zero at that point.

3.4. Inversion algorithm validation

To verify the accuracy and robustness of the inversion method, the data shown in Fig. 7 (b) are inverted from the numerically generated data by adding random errors [46].

$$T_{C,exp}(t) = T_{C,pred}(t) + \sigma_T \xi \quad (13)$$

where $T_{C,pred}(\tau)$ is the midplane temperature predicted by using the data presented in Fig. 7 (b), ξ is a normal distribution number with zero mean and unit standard deviation, while $\sigma_T = \delta T / 2.576$ is the standard deviation of measured temperature at the midplane for a δT measurement error at 99% confidence. Fig. 8 (a) presents the curves of the random measurement errors of $\delta T = 5$ K and 10 K, and the average effective thermal conductivity estimated (obtained by inversion algorithm) by ten independent inversions based on the experimental temperatures obtained by adding different random errors of δT is shown in Fig. 8 (b). The average value $\bar{k}_m$ is defined as

$$\bar{k}_m = \frac{1}{10} \sum_{i=1}^{10} k_{m,i} \quad (14)$$

where $k_{m,i}$ is the effective thermal conductivity obtained from each independent inversion results. The true and inversion values under different errors of δT are in good agreement, indicating that the inversion algorithm used in the present work has high accuracy and robustness.

4. Results and discussion

4.1. Effective thermal conductivity measured using the steady state method

In order to compare the results with the transient measurements, the effective thermal conductivity of ceramic fiber felts was measured in this paper using the steady state method. When the sample reaches the

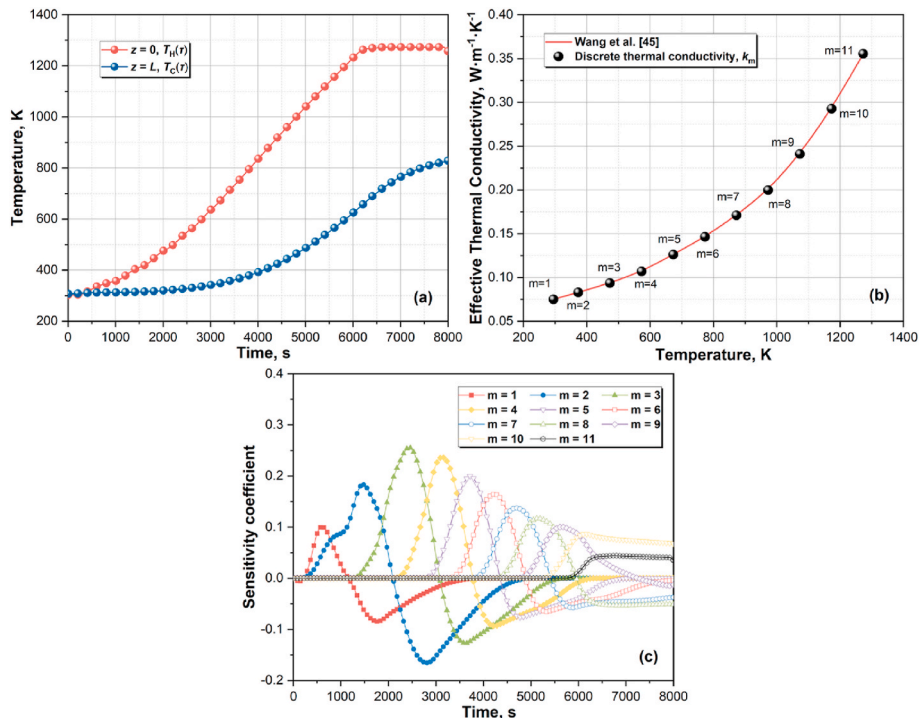


Fig. 7. Sensitivity analysis for k_m , (a) the boundary conditions, (b) the $k_{eff}(T)$ taken from the literature [45], and (c) sensitivity coefficients $\chi(k_m, t)$.

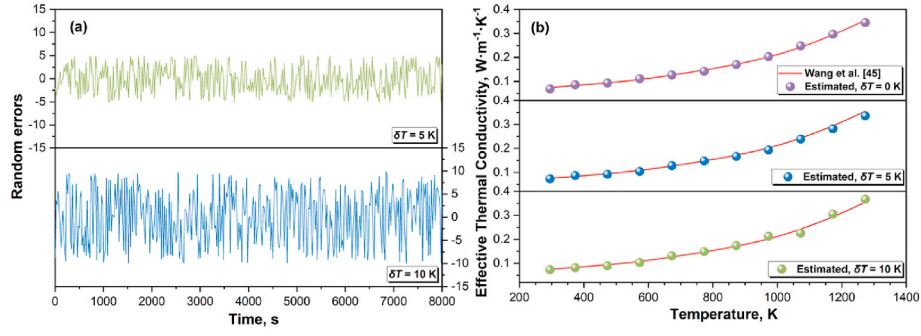


Fig. 8. Estimated $k_{eff}(T)$ from numerically generated temperature profiles added with a different measurement error of $\delta T = 0, 5$, and 10 K. (a) Measured error, (b) the estimated $k_{eff}(T)$.

steady state stage, the effective thermal conductivity of the sample can be calculated based on Fourier's law (Eq. (15)) by measuring the temperatures of the front and back surfaces of the sample as well as the heat flux through the sample, and the characteristic temperature is defined as $\bar{T} = (T_{H,S} - T_{L,S}) / 2$.

$$k_{eff}(\bar{T}) = \frac{q_t}{(T_{H,S} - T_{L,S})} \cdot L \quad (15)$$

where L is the sample thickness, q_t is the total heat flux along the sample thickness, $T_{H,S}$ and $T_{L,S}$ is the hot and cold surface temperature of the sample in steady state.

4.2. Comparison of measurement results between steady state and transient methods

The effective thermal conductivity results of the ceramic fiber felt

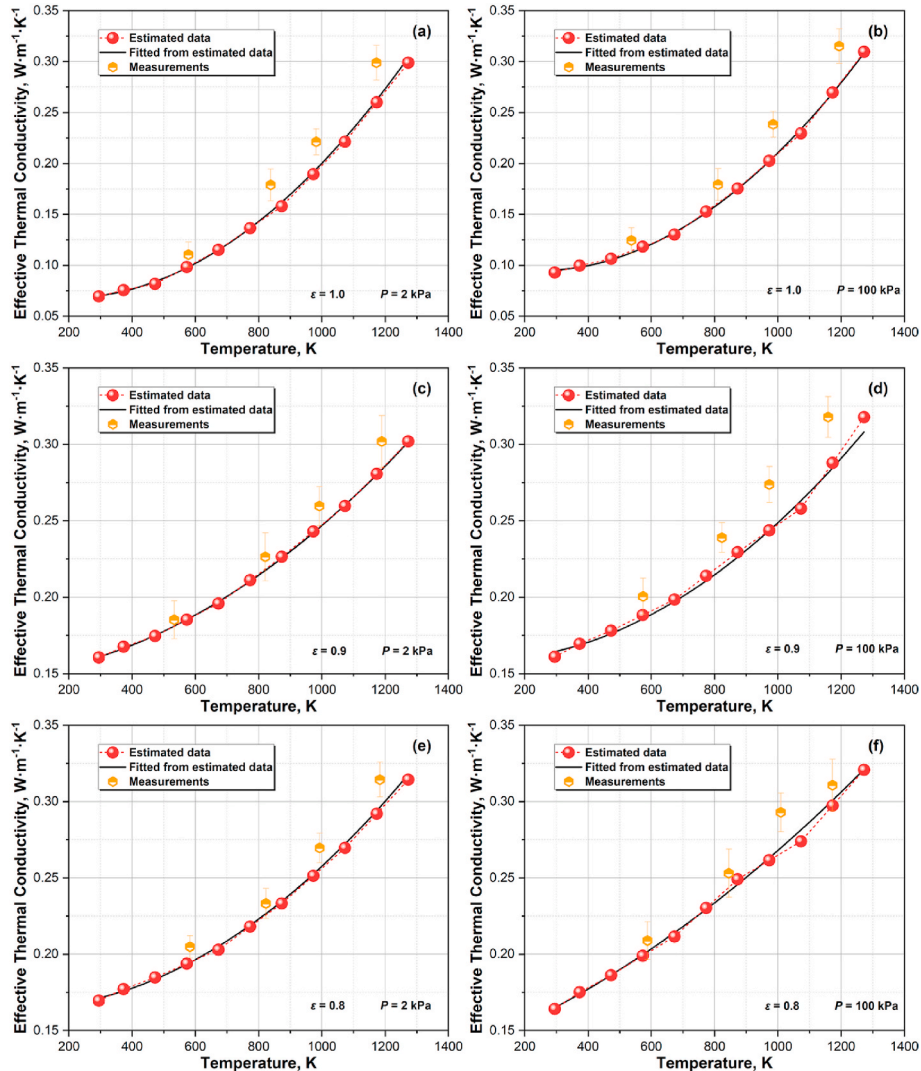


Fig. 9. Estimated and measured thermal conductivity of ceramic fiber felt as a function of temperature at (a) $p = 2$ kPa, and (b) $p = 100$ kPa.

with varying compression ratios and pressures are displayed in Fig. 9. The effective mean thermal conductivity measured through the steady state method, aligns with the inversion results' fitting curve. From the graph, it is evident that all curves demonstrate an increase in thermal conductivity with rising temperatures under different scenarios. As the temperature increases, radiation heat transfer gradually increases and becomes the primary method of heat transfer. The estimated data indicates that the ceramic fiber felt maintains strong thermal insulation performance at high temperatures, with values ranging from 0.069 to 0.321 $\text{W m}^{-1} \text{K}^{-1}$ between 295 K and 1273 K. Additionally, the ceramic fiber felt has compressible characteristics that make it a valuable component for reheat sealing structures, particularly in filling irregular or gap structures.

To ensure the accuracy of the thermal conductivity values obtained from the inversion, this work calculated the temperature data at the midplane of the sample using numerical simulation software and compared it with the experimental values. ANSYS Fluent 2022R1 was employed to solve the unsteady energy equations in a one-dimensional computational domain. The energy equation is discretized using a second-order upwind difference scheme. Convergence of the solution is achieved when the residual is less than 10^{-8} . The specific heat capacity and density of the sample are calculated using Eqs. (2) and (3), respectively. The thermal conductivity data for the samples is presented in Fig. 9. A user-defined function (UDF) is implemented to ensure that

the temperature boundary conditions match that used in the experiments. The initial temperature of sample is 290 K. The model employs a uniformly structured mesh with a mesh size of 0.1 mm, which was determined after passing the mesh independence test. The midplane experimental temperature of ceramic fiber felt versus the results from numerical simulation for various gas pressures and compression ratios are plotted in Fig. 10. Clearly, the simulations (the red line) align well with the corresponding experimental data (symbols). All predictions fall within the error margin of $\pm 3\%$ (within the red dotted line). Therefore, the identification inversion method used in this paper can precisely replicate the transient temperature within the sensed insulation.

4.3. Effect of compression ratio on effective thermal conductivity

In Fig. 11, a comparison is made between the effective thermal conductivity of ceramic fiber felt at different compression ratio, with the ambient pressure of 2 kPa and 100 kPa. The effective thermal conductivity of ceramic fiber felt shows a stepwise increase when the compression ratio changes from 1.0 to 0.9. By further compressing the ceramic fiber felt to 0.8, the value of effective thermal conductivity increases again. This significant change in thermal conductivity due to compression can be explained with two terms. On the one hand, compression reduces the porosity of ceramic fiber felt, resulting in a lower gas volume ratio and decreased thermal insulation performance.

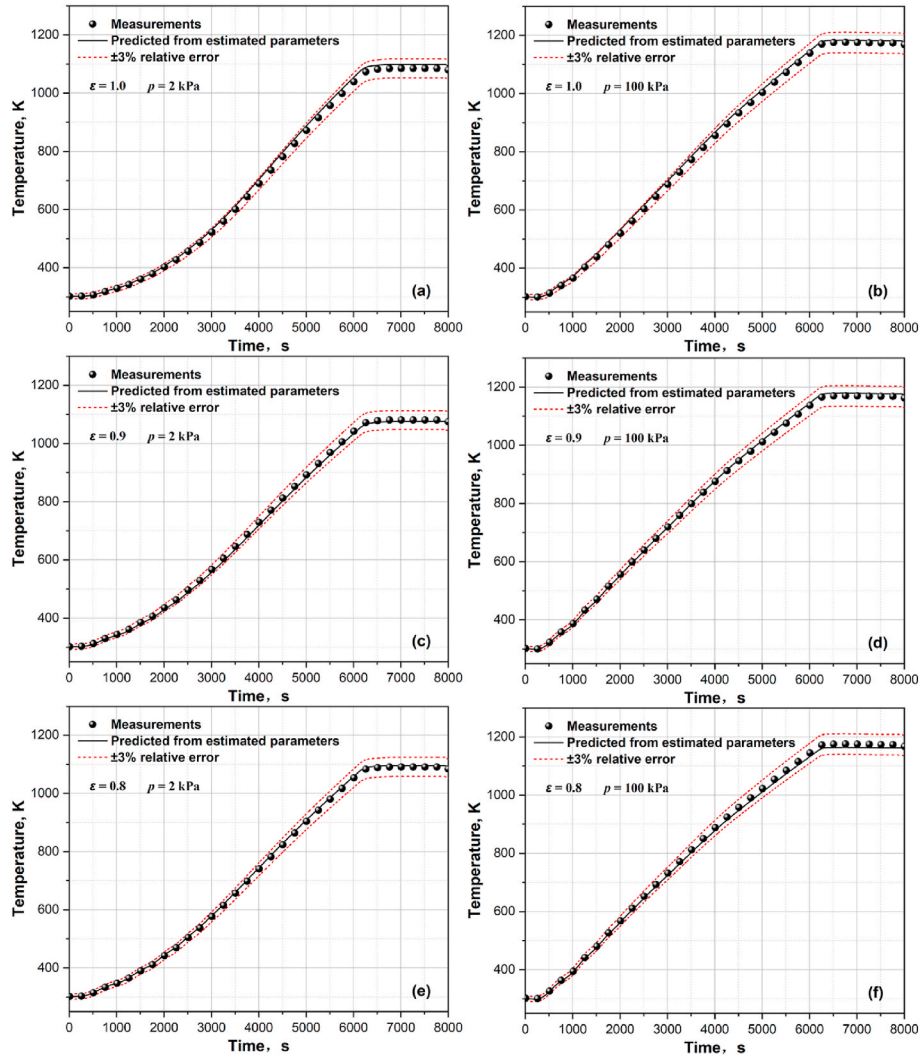


Fig. 10. Comparison of measured and numerically calculated values of the midplane transient temperature. (a) $\varepsilon = 1.0$, $p = 2$ kPa; (b) $\varepsilon = 1.0$, $p = 100$ kPa; (c) $\varepsilon = 0.9$, $p = 2$ kPa; (d) $\varepsilon = 0.9$, $p = 100$ kPa; (e) $\varepsilon = 0.8$, $p = 2$ kPa and (f) $\varepsilon = 0.8$, $p = 100$ kPa.

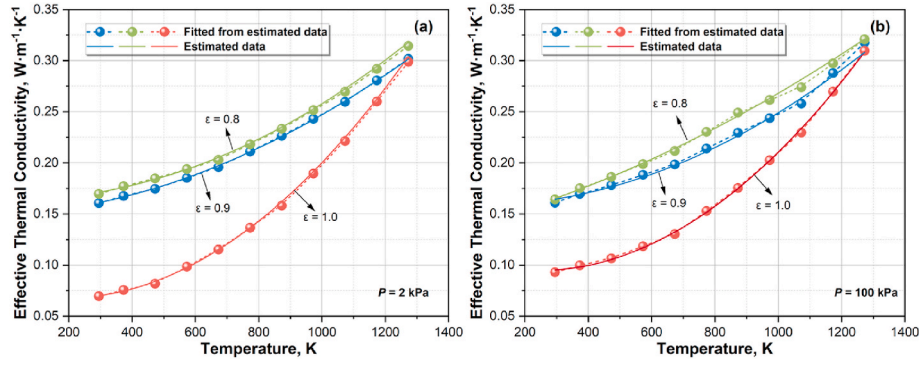


Fig. 11. Comparison of estimated effective thermal conductivity of ceramic fiber felt with different compression ratios at various gas pressure of (a) $p = 2$ kPa and (b) $p = 100$ kPa.

Typically, gas has a very low thermal conductivity. Moreover, increased contact points of ceramic fiber rods occur during the compression process. On the other hand, compression increases the sample's attenuation coefficient and inhibiting its radiation heat transfer capability, it also enhances thermal conductivity through the ceramic fiber rods. Nonetheless, at higher temperatures, radiation heat transfer increases rapidly. However, the ratio of thermal conductivity between gas and fiber rods inside the ceramic fiber felt to total heat transfer diminishes gradually with increasing temperature. At a temperature of 1273 K, the effective thermal conductivity of the ceramic fiber felt remains almost constant.

4.4. Effect of ambient pressure on effective thermal conductivity

Fig. 12 reports the effect of gas pressure on effective thermal conductivity of ceramic fiber felt with different comparison ratios. The thermal conductivity of the sample is affected by the ambient pressure in a compression rate-dependent manner. With increasing ambient pressure, more gas molecules in the sample pores are implicated in heat transfer, enhancing the sample's equivalent thermal conductivity when

uncompressed. As the sample is compressed, its internal pores shrink, causing a decrease in the molecular weight of the gas that can be held. This leads to a reduction in the heat transfer from the gas molecules within the sample. Therefore, there is negligible difference in thermal conductivity of the sample under two ambient pressures. But, once the sample is compressed even further, the difference in thermal conductivity between the two pressures re-emerges. Before each experiment, the interior of the furnace must be evacuated, followed by setting the ambient pressure to ensure that there is little to no gas present in the sample at the start of the process. We hypothesize that while the compression process reduces the pore volume within the sample, it also generates numerous confined spaces. At low temperatures, the thermal conductivity of the sample undergoes negligible changes due to the challenge that gas molecules in the environment encounter in penetrating the sample. This implies that the sample is shielded from external influences. As temperature increases, the fibers in the sample expand due to heat and compress against each other, causing the previously isolated space within the sample to connect with the external environment. Subsequently, as gas molecules enter the sample and participate in

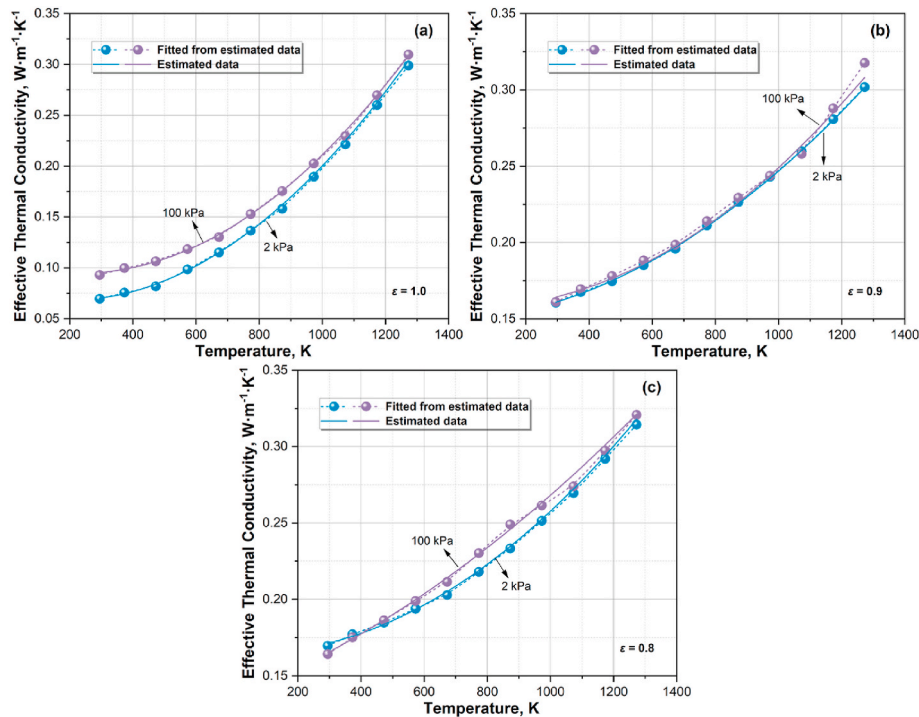


Fig. 12. Comparison of estimated effective thermal conductivity of ceramic fiber felt with a compression ratio of (a) $\epsilon = 1.0$, (b) $\epsilon = 0.9$ and (c) $\epsilon = 0.8$ at various gas pressure.

thermal conductivity, the effect of pressure on the sample's thermal conductivity gradually becomes apparent. Since the sample is challenging to compress further during the experiment, it is difficult to continue investigating the law of the two. However, it is conceivable that if the sample are to be compressed until it became a solid, gas would no longer be involved in heat transfer. Therefore, at this stage, the thermal conductivity of the sample would no longer be influenced by ambient pressure.

4.5. Error factor and model parameter uncertainty analysis

The thickness of the sample and the accuracy of temperature measurement have a significant impact on the measurement results of the thermal conductivity inversion. Therefore, it is essential to analyze the measurement uncertainties of these two factors. Moreover, we computed the uncertainties of the thermal conductivity obtained by steady state method and the specific heat capacity measurement results of the sample, and the results are shown in Table 1. There are two uncertainties, class A and class B, that are relevant in direct measurements. Class An uncertainty is calculated using the following equation:

$$U_A = t_p \cdot u_a \quad (16)$$

where t_p is a correction factor associated with the confidence interval p and the number of measurements n . u_a is the standard deviation of the arithmetic mean of the measurement data. The Class B uncertainty is calculated as:

$$U_B = \Delta_{ins}/C \quad (17)$$

where Δ_{ins} is the maximum allowable tolerance of instrument, $C = \sqrt{3}$. And the synthesized uncertainty in direct measurements is given by:

$$U = \sqrt{U_A^2 + U_B^2} \quad (18)$$

The thermal conductivity measured using the steady state method is calculated from other direct measurements, its synthetic uncertainty is determined according to the following equation:

$$U_k = \bar{k}_{ss} \cdot \sqrt{\left(\frac{\partial \ln k_{ss}}{\partial q_t}\right)^2 \cdot U(q_t)^2 + \left(\frac{\partial \ln k_{ss}}{\partial L}\right)^2 \cdot U(L)^2 + \left(\frac{\partial \ln k_{ss}}{\partial T_{H,S}}\right)^2 \cdot U(T_{H,S})^2 + \left(\frac{\partial \ln k_{ss}}{\partial T_{L,S}}\right)^2 \cdot U(T_{L,S})^2} \quad (19)$$

Likewise, the equation for the density of sample and the synthetic uncertainty equation for the density are:

$$\rho = \frac{M}{L \cdot L_a \cdot L_b} \quad (20)$$

$$U_\rho = \bar{\rho} \cdot \sqrt{\left(\frac{\partial \ln \rho}{\partial M}\right)^2 \cdot U(M)^2 + \left(\frac{\partial \ln \rho}{\partial L}\right)^2 \cdot U(L)^2 + \left(\frac{\partial \ln \rho}{\partial L_a}\right)^2 \cdot U(L_a)^2 + \left(\frac{\partial \ln \rho}{\partial L_b}\right)^2 \cdot U(L_b)^2} \quad (21)$$

where M is the quality of the sample; L , L_a , L_b are the thickness, length, and width of the sample, respectively.

The measurement error of L is evaluated as $\Delta L = 0.5$, and the intrinsic error of the thermocouples are estimated as $\Delta T = \max(2.2, 0.75\% \times T)$ K. Since temperature measurements can only be taken once, it is common to utilize Class B uncertainty as the final result's uncertainty since multiple measurements' outcomes do not reflect their

Table 1

Uncertainty of experimentally measured parameters.

Parameters	Measured value	U_A	U_B	U
L (mm), $\varepsilon = 1.0$	40	0.7623	0.0115	0.7623
L (mm), $\varepsilon = 0.9$	36	0.6409	0.0115	0.6410
L (mm), $\varepsilon = 0.8$	32	0.4964	0.0115	0.4965
L_a (mm)	300	0.9505	0.0115	0.9506
L_b (mm)	300	1.0334	0.0115	1.0335
M (g)	460.8	0.5550	0.0058	0.5550
ρ (g·cm ⁻³), $\varepsilon = 1.0$	0.128	—	—	0.0025
ρ (g·cm ⁻³), $\varepsilon = 0.9$	0.142	—	—	0.0026
ρ (g·cm ⁻³), $\varepsilon = 0.8$	0.160	—	—	0.0026
T (K)	1286	—	5.56	5.56
q (W·m ⁻²)	2408.3	—	41.7	41.7
c_p (J·kg ⁻¹ ·K ⁻¹)	1359.4	—	39.2	39.2
k_{ss} (W·m ⁻¹ ·K ⁻¹)	0.318	—	—	0.011

randomness. This also applies to the uncertainty surrounding the measurements of specific heat capacity. The Netzsch DSC 404 F3 and Fourier DaqPRO 5300 devices can maintain 5% and 3% maximum measurement error, respectively.

The thermal conductivity calculation in this paper, based on the inversion of transient temperature data, is subject to inherent limitations, as it is not a direct measurement or an accurate objective function. Therefore, this paper assesses the uncertainty of thermal conductivity data obtained from inversion by calculating the effect of uncertain parameters such as thickness, density, specific heat capacity, and measured temperature on the inversion results. Table 1 displays the parameter terms that correspond to the maximum uncertainty. The effect of temperature uncertainty on the inversion results has already been calculated in Section 3.4 and will not be repeated here. When calculating the influence of a parameter in the process, it is assumed that the remaining parameters are measured values without adding uncertainty. The change in thermal conductivity, obtained from the inversion after adding the effect of uncertainty, is calculated according to Eq. (22). Table 2 presents the thermal conductivity values with maximum error values and their corresponding temperature values and experimental pressures

obtained by inversion under the influence of uncertainties in the thickness, density, and specific heat capacity measurements of the ceramic fiber felt at different compression rates.

$$\omega = \frac{|k - k'|}{k} \times 100\% \quad (22)$$

where k is the thermal conductivity obtained by inversion without adding uncertainty to any parameters; k' is the thermal conductivity obtained by inversion with the addition uncertainty to a parameter.

Table 2

Error in thermal conductivity from inversion.

Set value	Thermal conductivity, $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$											
	$\varepsilon = 1.0$				$\varepsilon = 0.9$				$\varepsilon = 0.8$			
	p/kPa	T/K	k	$\omega/\%$	p/kPa	T/K	k	$\omega/\%$	p/kPa	T/K	k	$\omega/\%$
$L + U_L$	2	1273	0.311	3.9	2	1273	0.315	4.3	2	295	0.179	5.2
$\rho + U_\rho$	100	1273	0.319	2.9	100	295	0.161	3.3	2	1073	0.270	2.5
$c_p + U_{c_p}$	100	373	0.104	4.2	2	1173	0.281	4.8	100	295	0.164	3.6

5. Conclusion

In this study, we present an inversion method that allows for the calculation of the effective thermal conductivities of ceramic fiber felt at 11 temperature points ranging from 295 K to 1273 K. The method utilizes experimentally collected transient temperatures from the heated, cooled, and intermediate surfaces of the ceramic fiber felt while subjected to compression ratios of 0.8–1.0 and ambient pressures of 2 kPa and 100 kPa. This method boasts a high level of efficiency, accuracy, and reliability when compared to traditional steady state method measurements.

The inversion findings indicate that the thermal conductivity of the ceramic fiber mats ranges from 0.069 to 0.321 $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$. The effective thermal conductivity increases with increasing temperature. As the compression process elevates the heat conduction amongst the fiber rods in the ceramic fiber felt, the effective thermal conductivity of the ceramic fiber felt progressively increases at decreasing compression rates. After sustained compression, the ceramic fiber felt's porosity decreases, and the gas's heat transfer inside the pores decreases gradually. This leads to a stage of thermal conductivity that is not sensitive to ambient pressure changes. As the compression rate decreases further, there is a closed pore space in the ceramic fiber felt, preventing the entry of ambient gas. As temperature rises, the thermal expansion and deformation of the fiber rods lead to a loss of airtightness in the ceramic fiber felt, allowing gas to seep into the interior, thereby increasing the effective thermal conductivity of ceramic fiber felt. Moreover, the effect is amplified by higher ambient pressures. As the temperature increases, radiative heat transfer becomes increasingly dominant inside the

ceramic fiber felt. Consequently, the effective thermal conductivity of the sample is very small across different compression ratios and ambient pressures at 1273 K.

CRediT authorship contribution statement

Fei Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yinan Wang:** Validation, Methodology, Investigation. **Chuang Sun:** Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Pei Zhang:** Validation, Investigation. **Xinlin Xia:** Visualization, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Nomenclatures

c_v	specific heat capacity by volume [$\text{J}\cdot\text{m}^3\cdot\text{K}$]
φ	porosity
φ_0	initial porosity
c_p	specific heat capacity [$\text{J}\cdot\text{kg}\cdot\text{K}$]
ε	compression ratio
V	volume [m^3]
L	thickness [mm]
ρ	density [$\text{kg}\cdot\text{m}^3$]
p	pressure [Pa]
T	temperature [K]
τ	time [s]
k_c	thermal conductivity [$\text{W}\cdot\text{m}\cdot\text{K}$]
$\vec{q}$	radiation heat flux [$\text{W}\cdot\text{m}^2$]
σ	Stefan-Boltzman constant
n	refractive index
β	extinction coefficient
x	space variable [mm]
T_0	initial temperature [K]
N	number of temperature sampling points
ξ	normal distribution number with zero mean and unit standard deviation
σ_T	standard deviation
U	uncertainty

t_p	correction factor
n	number of measurements
u_a	standard deviation of the arithmetic mean
Δ_{ins}	maximum allowable tolerance of instrument
C	inclusion factor
M	quantity
L_a	length
L_b	width
ω	thermal conductivity error [%]

Subscripts

GA	genetic algorithm
C	compressed
H	high
L	low
M	midplane
t	total value
c	conduction
r	radiation
eff	effective value
R	Rosseland average
m	parameter number
exp	experimental
pred	predicted
S	specimen
A	class A
B	class B
p	confidence interval
SS	steady state
TS	transient state

Superscript

tr	transport
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Abbreviations

CNCs	cellulose nanocrystals
TPS	Transient planar source
AC	alternating current
SAC	SiO ₂ –Al ₂ O ₃ ceramic
TWA	temperature wave analysis
TC	thermocouple

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